Physical Chemistry (I)

Lecture 24

기말고사 안내 및 복습

기말고사 안내

- 시간: 6월 13일(월) 오후 3:00~4:00
- 장소: 화학관 B101호(홀수/짝수 학번 구분 없음)
- 계산기 불필요
- 신분증 지참

시험 범위

- 7장 “단순 혼합물”
- 8장 “화학 평형”
- 9장 “전기화학”
- 10장 “통계 역학”

시험 문제

총 일곱 문제

- 과제 문제 혹은 그 변형(3문제)
  - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 개념 설명 및 공식 유도(3문제)
  - 강의 자료 기준
- 보너스 문제(1문제)

Chapter 7

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Liquid mixture

Ideal solution

- Raoult's law

$$p_{\mathrm{A}} = x_{\mathrm{A}} p_{\mathrm{A}}^{*}$$

- Chemical potential

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT \ln x_{\mathrm{A}}$$

- $\Delta_{\mathrm{mix}} G = -T \Delta_{\mathrm{mix}} S$

Ideal-dilute solution

- Henry's law

$$p_{\mathrm{B}} = x_{\mathrm{B}} K_{\mathrm{B}}$$

- Chemical potential

$$\mu_{\mathrm{B}} = \mu_{\mathrm{B}}^{\ominus} + RT \ln x_{\mathrm{B}}$$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity

$$\mu_{\mathrm{A}} = \mu_{\mathrm{A}}^{*} + RT \ln a_{\mathrm{A}}$$

$$\mu_{\mathrm{B}} = \mu_{\mathrm{B}}^{\ominus} + RT \ln a_{\mathrm{B}}$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties

Concepts and Derivations in Chpt 7

- [C] Dalton's law
- [D] Mixing of perfect gases
- [C] Real gases: fugacity and fugacity coefficient
- [C] Mixing of liquids: liquid-gas equilibrium
- [C] Raoult's law and ideal solutions
- [D] Chemical potential of liquid
- [D] Mixing of ideal solution
- [D] Liquid and vapor compositions

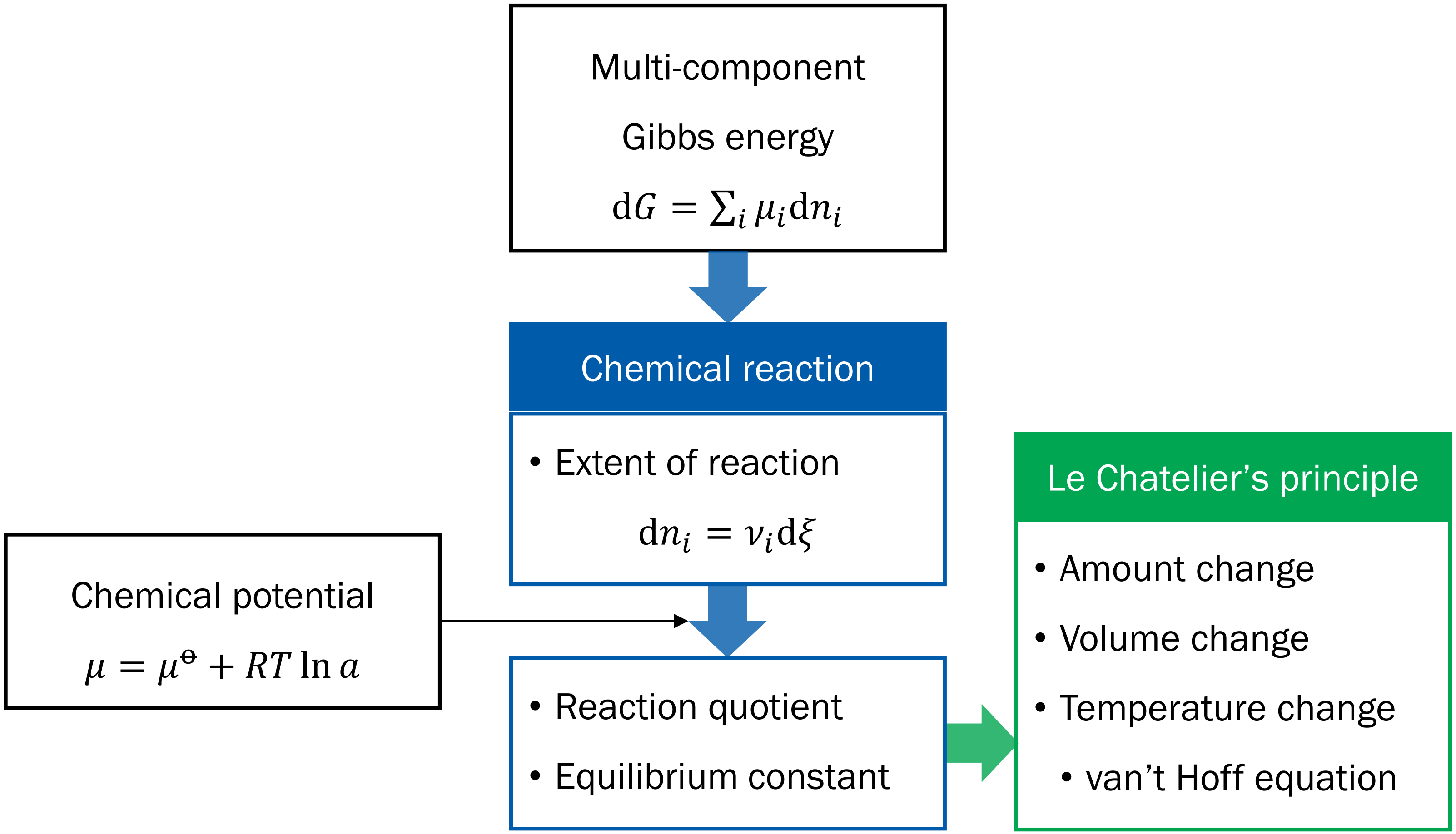
Concepts and Derivations in Chpt 7

- [C] Henry's law and ideal-dilute solutions
- [C] (Idealized) chemical potentials of solute and solvent
- [C] Bragg-Williams model
  - [D] Entropy of mixing
  - [D] Energy of mixing
  - [D] Helmholtz energy of mixing
- [C] Regular solutions
- [C, D] Excess function
- [C, D] Activity and activity coefficient
  - [D] Activity of Bragg-Williams model

Concepts and Derivations in Chpt 7

- [C] Phase diagrams of real solutions
  - [C] Distillation
  - [C] Liquid-gas equilibria: azeotropes
  - [C, D] Liquid-liquid equilibria: phase separation
- [C] Phase diagrams of liquid-solid mixtures
- [C, D] Colligative properties
  - Lowering of vapor pressure
  - Elevation of T_{b}, depression of T_{f}, solubility
  - Osmotic pressure

Chapter 8



Multi-component
Gibbs energy
$\mathrm{d}G = \sum_i \mu_i \mathrm{d}n_i$
Chemical reaction
• Extent of reaction
$\mathrm{d}n_i = \nu_i \mathrm{d}\xi$
Chemical potential
$\mu = \mu^{\ominus} + RT \ln a$
• Reaction quotient
• Equilibrium constant
Le Chatelier's principle
• Amount change
• Volume change
• Temperature change
• van't Hoff equation

Concepts and Derivations in Chpt 8

- [C] Chemical potential and activity
- [C, D] Reaction quotient and equilibrium constant
- [D] Activity of liquids and solids
- [D] Gas reactions
- [C, D] Le Chatelier's principle
  - Amount change
  - Volume change
  - Temperature change: van't Hoff equation

Chapter 9

$$\mathrm{d}G_{T,p} = \delta w_{\mathrm{add,max}}$$

$$\Downarrow$$

Electrochemical cell

- Cell potential

$$\Delta_{\mathrm{r}} G = -\nu F E_{\mathrm{cell}}$$

- Nernst equation

$$E_{\mathrm{cell}} = E^{\ominus}_{\mathrm{cell}} - \frac{RT}{\nu F} \ln Q$$

- Electrode potentials

Concepts and Derivations in Chpt 9

- [C] Electrochemical cells
- [D] Thermodynamics of cells
- [D] Nernst equation
- [D] Entropy and enthalpy of cell reactions
- [C, D] Electrode potentials

Chapter 10

$$S = k_B \ln W$$

Boltzmann factor

$$P(s) = \frac{1}{q} e^{-E(s)/k_B T}$$

- Partition function

$$q = \sum e^{-E(s)/k_B T}$$

Classical Stat Mech

Phase space

Partition function

$$q = \frac{1}{h^3} \iint e^{-E/k_B T}\, \mathrm{d}\vec{r}\,\mathrm{d}\vec{p}$$

- Many-particle PF

$$Q = \frac{1}{N!} q^N$$

Thermodynamics

- Helmholtz energy
- Internal energy

⋮

Perfect gas

- Partition function
- Helmholtz energy
- Pressure

Perfect gas law

Real gas

Equipartition theorem

- Partition function
- Internal energy

Concepts and Derivations in Chpt 10

- [D] Boltzmann factor
- [C] Partition function
- [C] Phase space
- [C] Partition function of classical system
  - [D] 1-dimensional spring
  - [D] N-particle partition function
  - [C] Indistinguishability

Concepts and Derivations in Chpt 10

- [D] Partition function and thermodynamics
  - Internal energy
  - Shannon entropy
  - Helmholtz energy
  - Other thermodynamic quantities
- [D] Perfect gas: perfect gas law
- [C] Real gas: van der Waals equation
- [D] Equipartition theorem